

JaeSeong Kim

Tel Aviv University
Department of Electrical Engineering

Email: jaeseongkim@mail.tau.ac.il
Homepage: jaeseongkim.com
[google scholar](#)

Research Interests	Image enhancement and restoration as inverse problems, with an emphasis on physics-aware reconstruction and learning-based priors. I am interested in improving mobile, automotive, and microscopy imaging through denoising, deblurring, super-resolution, and low-light/HDR enhancement, combining optimization with modern deep and generative methods.	
Education	Tel Aviv University	Tel Aviv, Israel
	<i>M. Sc. in Electrical Engineering (94.19/100)</i> Advisor: Prof. David Mendlovic	<i>April 2025</i>
	Tel Aviv University	Tel Aviv, Israel
	<i>B. Sc. in Electrical and Electronics Engineering (83.24/100)</i> • Leave of absence for mandatory military service (KATUSA), Oct 2017–Jun 2019	<i>July 2022</i>
Research Experience	Tel Aviv University & Samsung Corephotonics (Joint industrial Ms.c)	
	<i>Research Assistant (with Prof. David Mendlovic)</i> • Polarization imaging for mobile face anti-spoofing. Developed method achieving 10x lower error rate than RGB solutions. Simulated hybrid sensor architectures and recruited 64 participants for novel polarization face dataset.	<i>Oct 2022-Aug 2025</i>
	Tel Aviv University	
	<i>Research Assistant (with Prof. David Mendlovic)</i> Optics & Computer Vision Lab • Developed CNN-based crime classification system using skeleton tracking to analyze human movements in surveillance videos, classifying actions into 10 crime categories.	<i>Jul 2021-Jul 2022</i>
Publication & Patents	"On the Effectiveness of Sparse Linear Polarization Pixels for Face Anti-Spoofing", J. Kim , A. Pelz, M. Scherer and D. Mendlovic, <i>IEEE Sensors Journal</i> , 2025	
	• Presented at Israel Computer Vision Day 2025 (A. Pelz) • Accepted for oral presentation at Israel Machine Vision Conference (IMVC 2026) (A. Pelz)	
	"Sensors, systems and methods for compact face identification polarization cameras", M. Scherer, J. Kim , Patent WO2024161329A1, 2024	
	"Enhanced Face Anti-Spoofing using Angle of Linear Polarization", J. Kim , M.Sc. Thesis, Tel Aviv University, 2025	
Employment	Samsung Corephotonics	Tel Aviv, Israel
	<i>Image Quality Engineer</i> • Developed automated tool for quantifying EIS angular correction and motion	<i>Nov 2022-Apr 2025</i>

	blur, improving measurement accuracy by 25% and reducing analysis time from hours to 10 minutes • Tuned ISP parameters using Qualcomm Chromatix and measured image quality with IMATEST	
	Pit-In Global <i>Software Developer (internship)</i> • Created SQL database for location tracking by developing software for Raspberry Pi that leverages Bluez to collect Bluetooth signal strength indicators (RSSI) from customers' mobile devices	Seoul, Korea <i>Jul 2017-Sep 2017</i>
	CoderBunker <i>Community Manager</i>	Shanghai, China <i>Jan 2016-Mar 2016</i>
	Xinchejian(新车间) <i>Community Manager</i>	Shanghai, China <i>2015</i>
Presentations	Samsung Corephotonics Lecture Club "Face Anti-spoofing using full polarization characteristics" March 2025 Tel Aviv University Electrical Engineering Department Student Seminar "Enhanced Face Anti-spoofing using Angle of Linear Polarization" March 2025 Korean Embassy in Israel Research Seminar "Camera Technologies - Multimodal cameras & Image Quality Assessment" May 2024	
Skills	• Technical Skills (Image Processing, Computer Vision, Deep Learning) • Programming (Python, MATLAB, C/C++) • Tools / Software (IMATEST, OpenCV, PyTorch, Qualcomm Chromatix)	
Certifications	• Oracle Certified Professional, JAVA SE 6 Programmer • Deep Learning Specialization (by Andrew Ng, deeplearning.ai)	
Honors & Awards	• Certificate of Achievement, 8th U.S. Army • Certificate of Achievement, 2nd Battalion, 7th Infantry Regiment	<i>2018</i> <i>2018</i>
Additional Information	• Military Service: KATUSA (Korean Augmentation to the United States Army), Republic of Korea Army (Oct 2017 – Jun 2019) • Korean (Native), English (Native), Chinese (Native), Hebrew (Beginner) • Multicultural background with experience living in Korea, China (11 years), and Israel (6.5 years)	